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Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System

CSA1522-Cloud Computing and Big Data Analytics

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INTRODUCTION

- Infrastructure as a Service (IaaS) provides virtual computing resources over the internet.
- Virtual infrastructure includes Virtual Machines (VMs), storage, and networking.
- Auto-scaling automatically increases or decreases resources based on workload.
- It reduces cost while maintaining application performance.



OBJECTIVES

- To develop a cloud-based virtual infrastructure system.
- To automate VM provisioning.
- To implement auto-scaling based on resource usage.
- To improve scalability, availability, and cost efficiency.



LITERATURE REVIEW



Author / Study	Key Contribution	Limitation / Research Gap
VMware (Virtualization)	Introduced virtualization technology to create multiple virtual machines on a single physical server, improving resource utilization.	Focuses mainly on virtualization and does not provide intelligent auto-scaling.
Amazon Web Services (AWS EC2)	Provides Infrastructure as a Service (IaaS) with on-demand virtual machines, storage, and networking resources.	Resource provisioning requires proper scaling policies for optimal performance.
Microsoft Azure Auto Scale	Automatically adjusts computing resources based on CPU, memory, and workload demand.	Auto-scaling depends on predefined threshold values and may not predict future workloads.
Google Cloud Platform (GCP)	Offers scalable virtual infrastructure with load balancing and monitoring services.	Multi-cloud integration and advanced optimization are limited.
Proposed System	Proposed system uses OpenStack DevStack to provision virtual infrastructure while supporting future integration of monitoring, load balancing, and auto-scaling.	Aims to improve scalability, resource utilization, cost efficiency, and overall cloud performance.

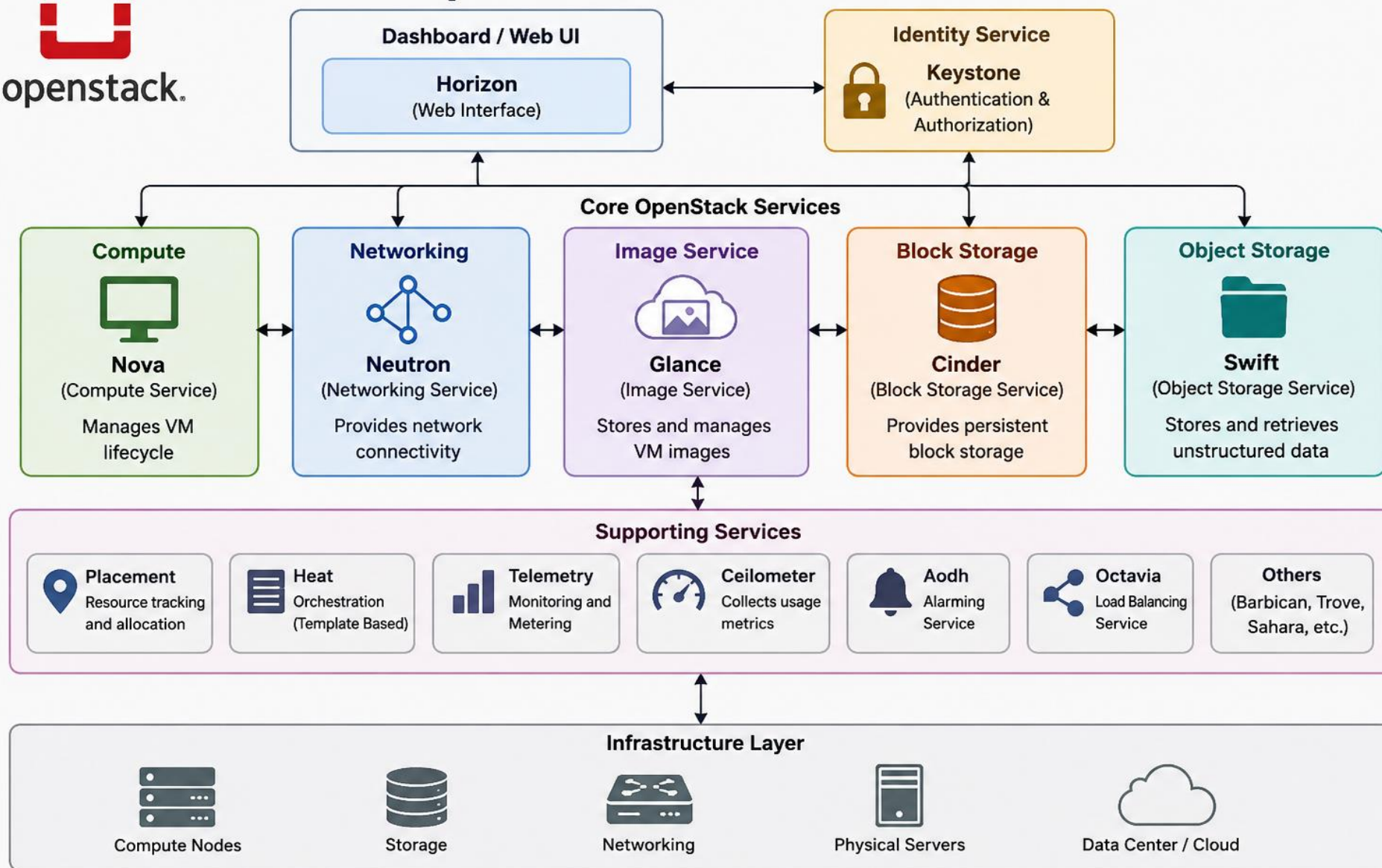


METHODOLOGY

Phase	Description	Tools/Technology
User Request	User logs in and requests a virtual machine (VM).	Horizon Dashboard
Cloud Controller	Authenticates the user and processes the provisioning request.	Cloud Management Platform
VM Provisioning	Creates and configures virtual machines with CPU, RAM, storage, and networking.	NOVA
Resource Monitoring	Continuously monitors CPU, memory, storage, and network utilization.	Horizon
Auto-Scaling	Automatically adds or removes VM instances based on workload and predefined thresholds.	Concept using Heat/Octavia
Service Delivery	Delivers scalable, highly available, and cost-efficient cloud services to users.	Load Balancer, Virtual Infrastructure



OpenStack Architecture





MODULE 1: VM Provisioning & Resource Allocation

- Receives user requests for virtual machines through the cloud portal.
- Authenticates users and verifies resource requirements.
- Automatically provisions Virtual Machines (VMs) using virtualization technology.
- Allocates CPU, RAM, storage, and network resources based on workload.
- Continuously manages and optimizes resource allocation for efficient utilization.

Figure 1.1: OpenStack Nova VM Provisioning Architecture.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack server list --all-projects
```

ID	Name	Status	Networks	Image	Flavor	Project ID
c448dca4-7c8a-4878-8ee8-dfb11ded7838	vm1	ACTIVE	private=10.0.0.24, fdc5:8689:3cae:0:f816:3eff:fe7f:c240	N/A (booted from volume)	m1.tiny	3a38c3c2ff4b4d53a80a37653a047f6a

Figure 1.2: VM1 Successfully Provisioned and Running.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack hypervisor list
```

ID	Hypervisor Hostname	Hypervisor Type	Host IP	State
a78710d8-fb4d-43e7-b8a2-d6d1a8454a22	LAPTOP-EDH45F4T.localdomain	QEMU	172.21.30.246	up



MODULE 2: Auto-Scaling & Load Balancing Engine

- Monitors CPU, memory, and network utilization.
- Uses predefined thresholds for scaling decisions.
- Demonstrates OpenStack's capability to integrate with Heat and Octavia for auto-scaling and load balancing.
- Distributes traffic using OpenStack networking architecture.
- Improves scalability and availability.

Figure 2.1: OpenStack Load Balancer Architecture.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack network list
```

ID	Name	Subnets
853ec43d-13ae-44a0-983e-92958b024417	shared	fa3dd3bf-d218-4fb7-95ac-fb3235d74c3f
8ee45ca0-4bbb-4b17-b3fa-cf4cfeaa0cf1	public	2e2de05f-d2cb-4fe7-b11e-d901b4f9ed85, 660731f8-7c2d-4211-ba27-84e22e247f66
d603f001-84c7-414e-979d-ecb1279479d1	private	4ef8330d-0686-4fee-9626-21cd4fedfaef, b7803708-d316-45f7-a396-5a750fb9c821

Figure 2.2: Network Configuration Supporting Load Balancing.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack router list
```

ID	Name	Status	State	Project	HA
06fff277-92f6-45e7-9168-b1a6e4639cf9	router1	ACTIVE	UP	3a38c3c2ff4b4d53a80a37653a047f6a	True

MODULE 3: Monitoring & Cloud Resource Management

- **Resource Monitoring** – Continuously monitors CPU, RAM, storage, and network usage of virtual machines.
- **Performance Tracking** – Checks the performance and health of active VMs in real time.
- **Threshold Detection** – Identifies when resource usage becomes too high or too low.
- **Resource Management** – Provides usage information to the auto-scaling engine for scaling decisions.
- **Cost Optimization** – Helps avoid unnecessary resource usage and improves overall cloud efficiency.

Figure 3.1: Horizon Monitoring Dashboard.

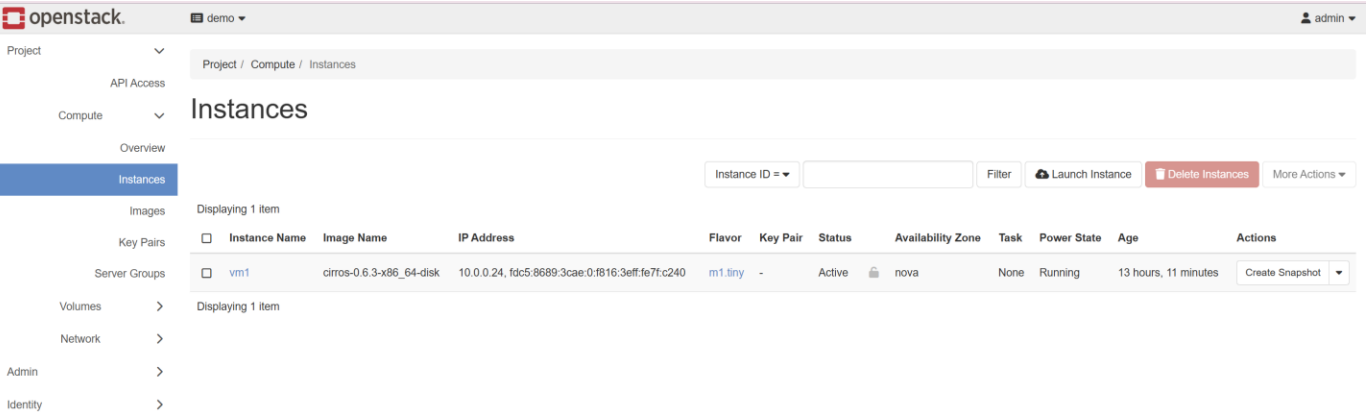


Figure 3.2: OpenStack Compute Services Status.

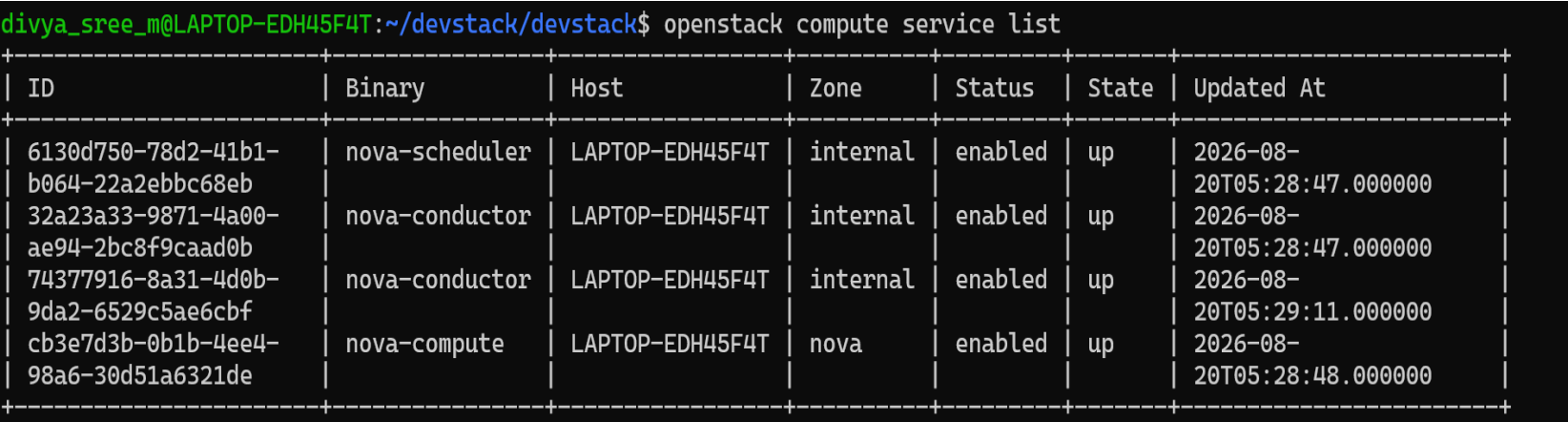


Figure 3.3: OpenStack Service Monitoring.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack server list --all-projects
```

ID	Name	Status	Networks	Image	Flavor	Project ID
c448dca4-7c8a-4878-8ee8-dfb11ded7838	vm1	ACTIVE	private=10.0.0.24, fdc5:8689:3cae:0:f816:3eff:fe7f:c240	N/A (booted from volume)	m1.tiny	3a38c3c2ff4b4d53a80a37653a047f6a



IMPLEMENTATION

- User authentication using Keystone
- VM provisioning using Nova
- Resource allocation (CPU, RAM, Storage)
- Network configuration using Neutron
- Storage attachment through Cinder
- Resource monitoring using Horizon Dashboard
- Successful deployment and activation of VM1



RESULTS & DISCUSSION

- Virtual infrastructure is provisioned quickly and efficiently.
- The deployed infrastructure successfully supports scalable VM provisioning and resource management.
- Resource utilization is optimized, reducing idle resources.
- Infrastructure costs are minimized through automatic scaling.
- The system provides high availability and reliable cloud services.



CHALLENGES & LIMITATIONS

- Cloud security and data privacy need continuous protection.
- Network latency may affect application performance.
- DevStack experienced an Out-of-Memory (OOM) issue in WSL, which was resolved by increasing swap memory and restarting MySQL and OpenStack services.
- Initial cloud setup and configuration are complex.
- Sudden workload spikes may cause temporary scaling delays.



FUTURE SCOPE

- Integrate AI for predictive auto-scaling decisions.
- Support multi-cloud deployment for better reliability.
- Use Docker and Kubernetes for container orchestration.
- Implement serverless computing for improved efficiency.
- Enhance security using Zero Trust and AI-based monitoring.



CONCLUSION

- Successfully deployed OpenStack DevStack on Ubuntu WSL.
- Auto-scaling improves resource utilization and performance.
- Load balancing ensures continuous service availability.
- Cloud resources are managed efficiently and cost-effectively.
- The solution provides a scalable, reliable, and flexible IaaS environment.



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Thank
you!